
Unifying Convolutional Dictionary Updates through a Grouped-Consensus Splitting

Anonymous

Abstract

The dictionary-update step is the computational bottleneck of convolutional dictionary learning (CDL): fixing the sparse codes, the filters solve a frequency-domain normal equation whose data-fidelity part has rank equal to the number of training images, so the rank-one Sherman-Morrison shortcut that makes sparse coding cheap does not apply. Several updates address this differently—an equality-constrained ADMM solved by iterated Sherman-Morrison or conjugate gradient, an ADMM consensus that gives each image its own filter copy, and an accelerated proximal-gradient update—and are usually presented as distinct algorithms. We show they are one algorithm. A single *grouped-consensus* splitting, in which the training images are partitioned into G groups and each group keeps a working copy of the filters coupled by a consensus average, has the equality update ($G=1$, one exact rank- N_b solve) and the consensus update ($G=N_b$, N_b rank-one solves) as its two extremes and the proximal-gradient update as its majorised limit; a handful of settings on one shared driver generate all of them. Instantiating this driver reproduces the reference implementations to a relative deviation below 10^{-6} at every iteration. The same driver, unchanged, expresses both standard multi-channel signal models—one multi-channel dictionary with a shared code, and one shared dictionary with per-channel codes—and runs both to convergence on colour data. Two consequences follow as algebra of the settings: the multi-channel update is exactly the grayscale update replicated across channels with a shared per-frequency factor, and single-channel-dictionary colour learning is exactly grayscale learning on the channel-images, quantifying the image/channel duality. Most usefully, the partition is a continuum: its interior—filter updates with intermediate group counts, absent from existing libraries and, to our knowledge, the literature—runs, converges, collapses onto the two reference endpoints when its group count is driven to either extreme, and occupies a genuinely cheaper per-iteration operating point than either. We give a selection map over the image-count/channel-count/correlation regime and a mechanism that reads cost and conditioning off the structure of the per-frequency system.

1 Introduction

Convolutional sparse representations model a signal as a sum of a few filters convolved with sparse coefficient maps, $\mathbf{s} \approx \sum_m \mathbf{d}_m * \mathbf{x}_m$ (Wohlberg, 2016b; Heide et al., 2015). Learning the filters from data—*convolutional dictionary learning* (CDL)—alternates a sparse-coding step that updates the coefficient maps with a *dictionary-update* step that updates the filters (Garcia-Cardona and Wohlberg, 2018a; Bristow et al., 2013). The dictionary update is the hard part. With the codes fixed, the filters minimise a least-squares data term subject to a support-plus-unit-norm constraint; in the DFT domain the normal equations decouple over frequency into independent systems, each a diagonal matrix plus a term of rank K , the number of training images (Šorel and Šroubek, 2016). Sparse coding, by contrast, faces a rank-one term per frequency and is dispatched by a single Sherman-Morrison solve.

The whole difficulty of the dictionary update is that this shortcut does not apply, and every published update is, at bottom, a different way of coping with the same rank- K structure.

The reference comparative study of these updates (Garcia-Cardona and Wohlberg, 2018a) catalogues an equality-constrained ADMM—whose inner system is solved by conjugate gradient (CG), iterated Sherman-Morrison (ISM), or spatial tiling—an ADMM consensus that gives each image a private filter copy so its system is rank-one, a frequency-domain “3D” variant, and an accelerated proximal-gradient (FISTA) update that avoids the linear solve altogether. Two of that study’s observations motivate this paper. First, two of the methods, derived independently and presented as distinct, turn out to be the *same* method in different coordinates (the “3D” update is consensus with the data term and constraint written in the DFT domain). Second, the choice among the methods is a choice among shared structural decisions—how the rank- K term is split, in which domain, with which inner solver—which is precisely why one can explain from the geometry of each linear system why one method beats another.

If several different-looking updates are secretly one structure, that structure should be writable as a single derivation whose settings generate the known methods—and whose *unfilled* settings generate new ones. And if a single derivation exists, it should carry over verbatim to the setting the reference study mostly set aside: real signals are multi-channel (colour, hyperspectral), and there is more than one way to represent channel information—a shared filter bank applied per channel with per-channel codes, a multi-channel filter bank with one shared code, or per-channel codes tied by an inter-channel penalty. These are genuinely different models, and it is not obvious which update suits which.

Contributions. We give one operator-splitting derivation of the convolutional dictionary update—a *grouped-consensus* problem in which the N_b data slots (one per training image, or one per image-channel pair) are partitioned into G groups, each keeping a working copy of the filters coupled by a consensus average—and a single ADMM driver for it (§3). From this one driver, by choosing a small set of settings:

1. we obtain the equality update ($G=1$), the consensus update ($G=N_b$), the CG and exact-ISM inner solves, and the proximal-gradient update (its majorised limit), and show each reproduces its reference implementation to a relative deviation $\leq 1.9 \times 10^{-7}$ at every iteration (§4);
2. we instantiate both standard multi-channel models and run them to convergence on colour images at the frozen regularisation, with no change to the driver (§5);
3. we fill the interior of the partition—updates with $1 < G < N_b$, which no released solver and, to our knowledge, no prior work provides—show they converge, and validate them by driving G to either extreme, where they collapse onto the ISM and consensus references to $\leq 1.2 \times 10^{-7}$ (§6);
4. we derive three equivalences as algebra of the settings—the multi-channel update as a shared-factor grayscale update, single-channel-dictionary colour learning as grayscale learning on the channel-images, and the “3D” update as consensus in a block-DFT basis—each confirmed numerically to machine precision (§7); and
5. we map which setting wins over which image-count/channel-count/correlation regime, and read the cost and conditioning of each off the per-frequency system (§8).

The point is not a new benchmark number: it is that one derivation reproduces the known updates, carries them to multi-channel data, and generates working methods between them, with the wins explained mechanistically.

2 Background: the dictionary-update subproblem

CDL over K training signals $\{\mathbf{s}_k\}$ with M filters $\{\mathbf{d}_m\}$ solves

$$\arg \min_{\{\mathbf{d}_m\}, \{\mathbf{x}_{m,k}\}} \frac{1}{2} \sum_k \left\| \sum_m \mathbf{d}_m * \mathbf{x}_{m,k} - \mathbf{s}_k \right\|_2^2 + \lambda \sum_{m,k} \|\mathbf{x}_{m,k}\|_1 \quad \text{s.t.} \quad \|\mathbf{d}_m\|_2 = 1, \quad (1)$$

non-convex jointly but convex in each block, and is minimised by alternating a sparse-coding step (update $\{\mathbf{x}_{m,k}\}$ by convolutional basis pursuit denoising, CBPDN) with a dictionary-update step (update $\{\mathbf{d}_m\}$), coupled through auxiliary variables with a single inner iteration of each before alternating (Garcia-Cardona and Wohlberg, 2017). We hold the functional (1), the constraint, the coupling, and the sparse coder ($\lambda=0.1$) fixed throughout; our object is the dictionary update.

Table 1: Coordinates of the grouped-consensus driver and the values used in this paper. Only G , the inner solver, and the channel model differ between the methods in Table 2; the driver code is one shared body.

Coordinate	Values	Role
group count G	$1, \dots, N_b$	rank $ \mathcal{I}_g $ of each per-frequency system
inner solver	ISM / SM / CG / linearise	how (4) is (approximately) inverted
slot basis	spatial / block-DFT	domain in which the coupling is written
channel model	mc-dict / sc-dict	shape of the filter bank (C channels vs. 1)
acceleration	none / Nesterov	ISTA vs. FISTA in the forward-backward limit

Writing X_k for the linear operator that convolves the fixed codes of image k with the filter bank $\mathbf{d}$ (so $X_k \mathbf{d} = \sum_m \mathbf{x}_{m,k} * \mathbf{d}_m$) and stacking $X = (X_0^\top X_1^\top \dots)^\top$, $\mathbf{s} = (\mathbf{s}_0^\top \mathbf{s}_1^\top \dots)^\top$, the update is

$$\arg \min_{\mathbf{d}} \frac{1}{2} \|X\mathbf{d} - \mathbf{s}\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{d}), \quad (2)$$

where $\iota_{C_{\text{PN}}}$ is the indicator of the support-plus-unit-norm set; its proximal map crops each filter to its support, (optionally) removes its mean, zero-pads, and normalises to unit norm. The gradient of the smooth term is $X^H(X\mathbf{d} - \mathbf{s})$, and $X^H X = \sum_k X_k^H X_k$ is, per DFT frequency, a diagonal matrix plus a term of rank K : the sum of K rank-one outer products, one per image. This rank- K term is the entire source of difficulty, and the known updates differ only in how they handle it: solve the rank- K system exactly (ISM, $\mathcal{O}(K^2)$ per frequency) or iteratively (CG); split it into K rank-one systems coupled by a consensus (each an $\mathcal{O}(1)$ Sherman-Morrison solve); or never form it and take gradient steps (FISTA).

3 One splitting for the dictionary update

A grouped-consensus master problem. Let there be N_b data slots. For grayscale or a multi-channel filter bank with a shared code, $N_b = K$ (one slot per image); for a single-channel filter bank on C -channel data, $N_b = CK$ (one slot per image-channel pair, §5). Partition the slots into G groups $\{\mathcal{I}_g\}_{g=1}^G$, give each group its own working copy $\mathbf{d}_g$ of the filter bank, and tie all copies to a single constrained variable $\mathbf{g}$:

$$\arg \min_{\{\mathbf{d}_g\}, \mathbf{g}} \sum_{g=1}^G \frac{1}{2} \sum_{k \in \mathcal{I}_g} \|X_k \mathbf{d}_g - \mathbf{s}_k\|_2^2 + \iota_{C_{\text{PN}}}(\mathbf{g}) \quad \text{s.t.} \quad \mathbf{d}_g = \mathbf{g} \quad \forall g. \quad (3)$$

Problem (3) is a consensus problem over G blocks whose block g carries the data of exactly the images in $\mathcal{I}_g$. Its scaled-ADMM iteration, with penalty ρ , over-relaxation α , and scaled duals $\mathbf{u}_g$, is our single driver:

$$\mathbf{d}_g \leftarrow \left(\sum_{k \in \mathcal{I}_g} X_k^H X_k + \rho I \right)^{-1} \left(\sum_{k \in \mathcal{I}_g} X_k^H \mathbf{s}_k + \rho(\mathbf{g} - \mathbf{u}_g) \right), \quad g = 1, \dots, G, \quad (4)$$

$$\mathbf{g} \leftarrow \text{prox}_{\iota_{C_{\text{PN}}}} \left(\frac{1}{G} \sum_g (\alpha \mathbf{d}_g + (1-\alpha)\mathbf{g} + \mathbf{u}_g) \right), \quad (5)$$

$$\mathbf{u}_g \leftarrow \mathbf{u}_g + \alpha \mathbf{d}_g + (1-\alpha)\mathbf{g} - \mathbf{g}. \quad (6)$$

Step (4) is the group solve: per frequency, a diagonal matrix plus a term of rank $|\mathcal{I}_g|$. Step (5) averages the group estimates and projects onto the constraint. Step (6) is the dual ascent. Every operation in (4) to (6) is written once, parameterised by the partition and by how (4) is solved.

Settings. The behaviour of the driver is fixed by a few coordinates (Table 1): the partition granularity G (equivalently the group rank $\lceil N_b/G \rceil$); the inner solver for (4); the change of basis over the slot axis; the channel model, which is nothing but the shape of the initial filter bank; and—orthogonal to all of these—whether the update takes the ADMM form above or its majorised forward-backward limit, defined next. A scaled-dual initialisation convention (§4) is held to each reference and is not a modelling choice.

Design space: representation $\times$ group-axis cell

grayscale	recovered	new	recovered	recovered	equiv.
multi-channel dict (shared map)	recovered	new	recovered	recovered	predicted
single-channel dict (per-channel maps)	recovered	new	recovered	recovered	predicted
	G=1 exact (ISM/CG)	1<G<N _b grouped	G=N _b consensus	forward-backward (FISTA)	block-DFT basis (3D)
	recovered vs SPORCO	new (un-instantiated) cell	equivalence re-derived		predicted, not run

Figure 1: The design space as a map: each representation (row) against the group-count/inner-solver cell (column). Blue cells are recovered against SPORCO (§4, §5); orange cells are the new grouped interior (§6); the aqua cell is the block-DFT “3D” equivalence demonstrated for grayscale (§7); grey cells are predicted by the same coordinates but not run here.

The two extremes. With $G=1$ there is a single group holding all N_b slots, so (4) is one exact rank- N_b solve $(\sum_k X_k^H X_k + \rho I)\mathbf{d}_1 = \sum_k X_k^H \mathbf{s}_k + \rho(\mathbf{g} - \mathbf{u}_1)$, the average in (5) is trivial, and the iteration is *term-for-term* the equality-constrained ADMM update: ISM if (4) uses iterated Sherman-Morrison, CG if it uses conjugate gradient. With $G=N_b$ each group holds one slot, every (4) is a rank-one Sherman-Morrison solve, and the iteration is the ADMM consensus update. The interior $1 < G < N_b$ is new (§6).

The forward-backward limit. Setting the inner solver to *linearise* replaces the exact solve (4) by a single majorisation step. Majorising the smooth term at the current iterate by $\frac{L}{2} \|\mathbf{d} - \mathbf{d}^{(i)}\|_2^2$ with $L \geq \|X^H X\|$ cancels the Gram matrix and turns the filter update into a gradient step followed by the constraint prox,

$$\mathbf{g}^{(i+1)} = \text{prox}_{\iota_{\text{CPN}}}(\mathbf{d}^{(i)} - \frac{1}{L} X^H (X \mathbf{d}^{(i)} - \mathbf{s})), \quad (7)$$

with the gradient $X^H (X \mathbf{d} - \mathbf{s})$ formed in the DFT domain; Nesterov extrapolation of $\mathbf{d}^{(i)}$ makes (7) the FISTA update (Beck and Teboulle, 2009). No linear system is ever formed, which is exactly why the proximal-gradient corner sidesteps the rank- N_b obstruction. Thus the ADMM branch (4) to (6) and the forward-backward branch (7) are two faces of the same operator-splitting family: the second is the first with the fidelity solve replaced by its majoriser.

Why this is one derivation, not a dispatcher. The evidence is the interior. A dispatcher that branched to four hand-written solvers could never produce a working method from a branch nobody wrote; the grouped update with $1 < G < N_b$ arises with *no* new code—the same (4) to (6) with a different partition—and its two degenerate limits fall back exactly onto the two reference endpoints (§6). We reuse only the frozen numerical kernels from the reference library SPORCO (Wohlberg, 2017): the convolutional-representation indexing and the constraint prox, the FFT wrappers, and the three diagonal-plus-low-rank solve primitives (iterated Sherman-Morrison, single Sherman-Morrison, conjugate gradient) that are the values of the inner-solver coordinate. Everything that assembles the per-group systems, forms the consensus, and drives the iteration is the single body above.

4 The known updates are settings of one driver

Figure 1 maps the design space. We now instantiate the driver and check it against SPORCO’s own solvers on identical inputs—the same seed, initial filters, images, λ , penalty/step, iteration budget, and inner-solver configuration (CG tolerance and cap; FISTA step $L=14K$ and Nesterov momentum). All four SPORCO families are numerically deterministic under these settings, so a correct instantiation must track them tightly. We measure the relative deviation of the CDL functional (1)—evaluated identically for both, with unit-norm filters—at every logged iteration, and the relative Frobenius error of the converged filters. The scaled dual is initialised with each reference family’s convention (the equality family uses $\mathbf{u}^0 = \mathbf{g}^0$, the consensus family $\mathbf{u}^0 = \mathbf{g}^0 / \rho$); this perturbs only the first iterate and not the fixed point (§9), and is disclosed as a parity setting, not a modelling choice.

Table 2: Each method is one setting of the driver of §3. Grayscale at $K=8$, $N=32 \times 32$, $M=16$, 40 iterations; colour at $K=5$, $C=3$, $N=32 \times 32$, $M=16$, all at $\lambda=0.1$. “Recovery deviation” is the maximum relative deviation of the CDL functional from the matched SPORCO run over all iterations; for the interior cells (*), which have no external reference, it is the degenerate-collapse check (§6). Converged functionals are comparable within each representation block (shared configuration).

Method	Representation	Settings	Recovery dev.	Conv. func.
ISM	grayscale	$G=1$, exact (ISM)	8.3×10^{-8}	26.90
CG	grayscale	$G=1$, CG	7.6×10^{-8}	26.90
Consensus	grayscale	$G=N_b$, rank-1 (SM)	1.3×10^{-7}	29.02
FISTA	grayscale	linearise, Nesterov	1.4×10^{-7}	28.49
Grouped consensus *	grayscale	$G=4$, $1 < G < N_b$, consensus	collapse $\leq 1.3 \times 10^{-7}$	28.16
ISM	multi-channel dict.	$G=1$, exact (ISM)	1.3×10^{-7}	57.55
Consensus	multi-channel dict.	$G=N_b$, rank-1 (SM)	1.9×10^{-7}	58.82
ISM	single-channel dict.	$G=1$, exact (ISM)	1.2×10^{-7}	67.57
FISTA	single-channel dict.	linearise, Nesterov	1.6×10^{-7}	69.64
Grouped consensus *	single-channel dict.	$G=3$, $1 < G < N_b$, consensus	collapse $\leq 1.2 \times 10^{-7}$	68.39

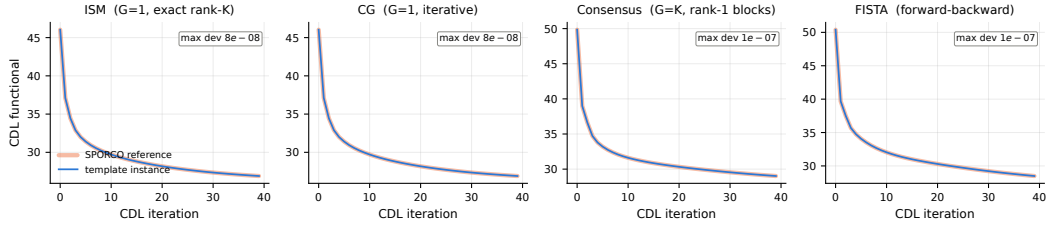


Figure 2: Each grayscale update, instantiated from the grouped-consensus driver (blue), lies on top of its SPORCO reference (orange) at every CDL iteration; the maximum relative deviation is annotated. ISM and consensus are $G=1$ and $G=N_b$ of the same driver; CG and FISTA change only the inner solver.

Table 2 reports the result and Fig. 2 overlays the trajectories. All four grayscale families are reproduced to a relative functional deviation below 1.9×10^{-7} and a converged-filter Frobenius error below 3×10^{-6} , one to two orders tighter for the two exact DFT solves. The match is essentially machine precision in single-precision arithmetic; the driver is not approximating the reference updates but computing the same iterates. The two exact solves (ISM, consensus) and two different solver families (CG, FISTA) are all recovered, from one body of code, by changing G and the inner solver.

5 Multi-channel signals are settings of the same driver

For C -channel signals two representations are standard (Wohlberg, 2016a; Garcia-Cardona and Wohlberg, 2018a). A *multi-channel dictionary* (mc-dict) uses C -channel filters $\mathbf{d}_{m,c}$ with a single shared code $\mathbf{x}_{m,k}$; a *single-channel (shared) dictionary* (sc-dict) uses one-channel filters with per-channel codes $\mathbf{x}_{m,k,c}$. In the driver these are nothing but the shape of the initial filter bank: C channels for mc-dict (so $N_b = K$), one channel for sc-dict. In the sc-dict case a single-channel filter must explain C channels of every image, so the channel index behaves in the update exactly like an image index; the driver folds the C channels onto the slot axis, giving $N_b = CK$. No other line changes.

Both representations run to convergence on colour images at the frozen $\lambda=0.1$ ($N=64 \times 64$, $K=8$, $M=64$, $C=3$, 100 iterations, 2 seeds), each compared against a matched SPORCO reference for that representation run at the *same* λ (not the shipped $\lambda=0.2$ colour examples, which minimise a differently weighted functional). The multi-channel dictionary reduces the functional to between 0.32 and 0.33 of its initial value and the single-channel dictionary to between 0.27 and 0.28 (Fig. 3); both sit within 9.3×10^{-8} (mc-dict) and 0 (sc-dict) of the matched reference at every iteration. The forward-

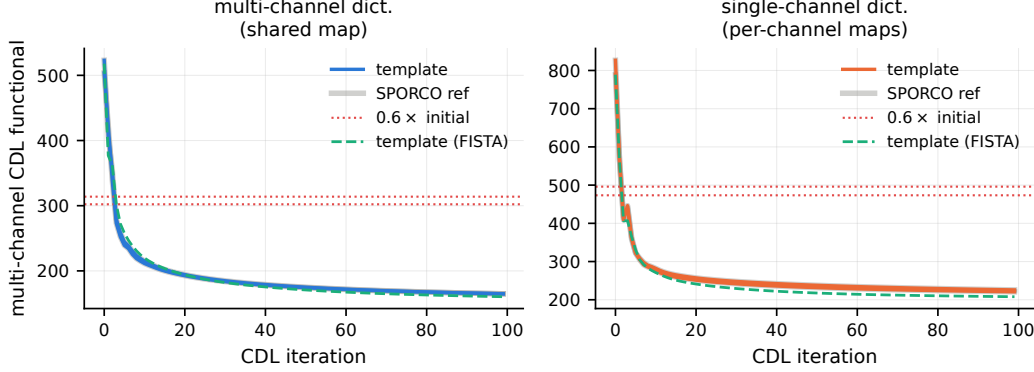


Figure 3: Both multi-channel representations, instantiated from the same driver, converge on colour images at $\lambda=0.1$ (two seeds each), tracking their matched SPORCO references (grey) and falling well below $0.6\times$ their initial functional (red dotted). The forward-backward instance (dashed aqua) converges too.

backward corner converges on colour as well (dashed). Reproducing these two representations on colour is baseline work carried by the reviewed study’s own Sec. VI (Garcia-Cardona and Wohlberg, 2018a); our point is that they cost *no new derivation*—they are two settings of the driver of §3—which is what lets us treat their equivalences (§7) and their interior (§6) uniformly with the grayscale case.

6 Methods between the extremes

Existing solvers offer only the two ends of the partition: one rank- N_b system (equality ADMM) or N_b rank-one systems (consensus). The driver’s partition is a continuum, and its interior $1 < G < N_b$ —each group an exact solve of an intermediate-rank system, the groups coupled by a consensus average—is, to our knowledge, absent from released CDL software and from the literature. It is a concrete, runnable method: partition the images into G groups, solve each group’s rank- $\lceil N_b/G \rceil$ system by ISM, average and project.

It runs and converges. On grayscale data with $K=16$ images, the interior counts $G \in \{2, 4, 8\}$ all reduce the CDL functional monotonically to within a fraction of a percent of the two endpoints (Fig. 4, left). On colour data with a single-channel dictionary, grouping the $N_b=12$ image-channel slots—along images or along channels, a two-parameter family opened by the duality of §7—likewise converges.

It collapses onto both references. The decisive check that the interior is a genuine specialisation of one derivation, not a wrapper, is that driving its group count to either extreme reproduces a reference the interior never targeted. From the *same* grouped code, $G=1$ matches SPORCO’s ISM to 8.5×10^{-8} and $G=N_b$ matches SPORCO’s consensus to 1.2×10^{-7} (grayscale); in the colour setting, one group matches ISM to 8.4×10^{-8} and N_b groups match consensus to 1.1×10^{-7} (Fig. 4, right). Only the partition changes between these runs. A dispatcher over separate solvers cannot produce this behaviour, because the interior corresponds to no pre-written branch.

It is a distinct operating point. The interior is not a curiosity between two better endpoints. Because ISM’s per-frequency cost grows as the square of the group rank while the consensus average grows linearly in the number of groups, the total per-iteration cost is non-monotonic in G : on $K=12$ images the cheapest per-iteration update is $G=4$ (9.34 ms), below both the $G=1$ endpoint (15.05 ms) and the $G=N_b$ endpoint (12.58 ms), while its convergence in outer iterations sits between them (§8). The partition exposes a cost/conditioning trade-off that neither endpoint alone reveals.

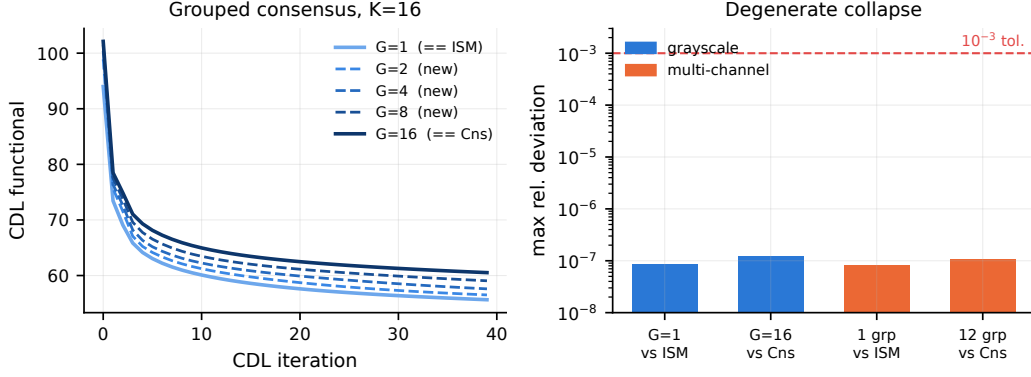


Figure 4: **Left:** grouped-consensus updates for $G \in \{1, 2, 4, 8, 16\}$ on $K=16$ images. The endpoints $G=1$ (ISM) and $G=16$ (consensus) are solid; the interior cells (dashed), which no released solver provides, converge between them. **Right:** driving the interior driver’s group count to either extreme reproduces the SPORCO ISM and consensus references to $\sim 10^{-7}$, far inside the 10^{-3} tolerance (dashed), in both the grayscale and multi-channel settings.

7 Equivalences that fall out of the settings

A genuine derivation should make known equivalences appear as algebra of its settings rather than as separate proofs. Three do.

The multi-channel update is a shared-factor grayscale update. For an mc-dict model the code $\mathbf{x}_{m,k}$ carries no channel index, so in (4) the operator $\sum_k X_k^H X_k + \rho I$ is *identical* for every channel c ; only the right-hand side $\sum_k X_k^H \mathbf{s}_{k,c}$ differs. The C -channel update is therefore C grayscale-shaped updates sharing one per-frequency factorisation—one factor, C back-substitutions—so the multi-channel dictionary update is no more expensive than grayscale, recovering Eq. (90) of (Garcia-Cardona and Wohlberg, 2018a) as a property of the setting. Numerically, the vectorised C -channel solve and a loop of C single-channel solves that reuse the same operator agree to 0 (the operator’s channel dimension is 1, i.e. channel-independent by construction).

Single-channel-dictionary colour learning is grayscale learning on the channel-images (the $K \leftrightarrow C$ duality). Folding the C channels onto the slot axis makes the sc-dict dictionary update on (C, K) colour data *identical* to a grayscale update on CK images: the internal coefficient-operator spectra coincide, and the two updates produce the same filter sequence. With $C=3, K=4 \rightarrow 12$ image-channel slots the two updates agree to 0 over 25 iterations. This makes concrete the duality noted in (Garcia-Cardona and Wohlberg, 2018a): the channel count C plays for the dictionary update the rank role the image count K plays, and—dually—for sparse coding the roles swap. The selection map of §8 uses this correspondence.

The “3D” update is consensus in a block-DFT basis. Writing the data term and the equality constraint in a real orthonormal basis over the slot axis leaves the objective invariant (Parseval): at any filter bank the data-fidelity value in the spatial and block-DFT bases agree to 0. The specific block-DFT that diagonalises the block-circulant “3D volume” operator $\begin{pmatrix} X_0 & X_1 \\ X_1 & X_0 \end{pmatrix}$ into the per-slot operators $X_0 \pm X_1$ —verified to 1.8×10^{-7} —turns (3) at $G=N_b$ into the frequency-domain “3D” update. Thus the “3D” method is the consensus setting of the driver read in one more basis, the change of the slot-basis coordinate, recovering the reviewed study’s own unification as a coordinate flip. (Because the constraint set is non-convex, the two bases follow different paths to possibly different stationary points; the equivalence is of the problem and its class, as in the original derivation, not of the iterates.)

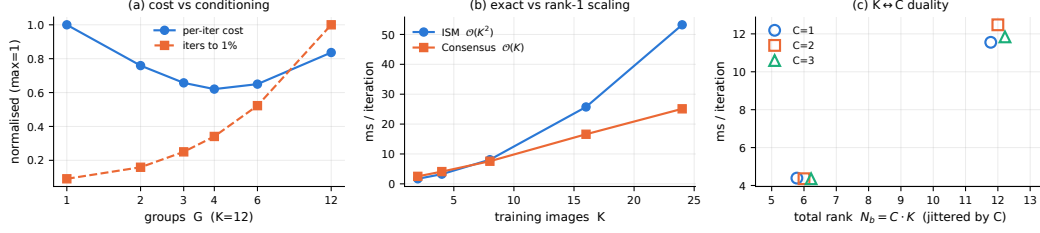


Figure 5: **(a)** On $K=12$ images the per-iteration cost is non-monotonic in the group count G (minimum at $G=4$), while the outer iterations to reach 1% of the best functional rise with G : a cost/conditioning trade-off. **(b)** Exact ISM scales as $\mathcal{O}(K^2)$ and rank-one consensus as $\mathcal{O}(K)$ per iteration; ISM wins for small K (better constant) and consensus for large K (better exponent), crossing near $K=8$. **(c)** Update cost depends only on the total rank $N_b=CK$, not on how it splits between images and channels: the $C=1, 2, 3$ points coincide at each N_b , quantifying the $K \leftrightarrow C$ duality.

8 Which setting wins where

The settings are not interchangeable, and the driver lets us say why from the per-frequency system rather than from timing alone. An exact rank- r solve by ISM costs $\mathcal{O}(r^2)$ per frequency; splitting the rank into G groups of rank $\lceil N_b/G \rceil$ and averaging costs $\mathcal{O}(G(N_b/G)^2) + \mathcal{O}(G) = \mathcal{O}(N_b^2/G) + \mathcal{O}(G)$ per frequency, so linear-algebra cost falls with G while the averaging and the conditioning penalty rise. Fig. 5 reads this off directly.

A selection map. (i) *Number of images K .* For a single exact solve, ISM’s $\mathcal{O}(K^2)$ per iteration beats consensus’s $\mathcal{O}(K)$ in constant for small K but loses for large K ; on our data the crossover is near $K=8$ (53.22 ms vs. 25.11 ms at $K=24$, Fig. 5b). (ii) *The interior.* When neither endpoint is ideal— K large enough that a full ISM solve is costly, but consensus’s rank-one conditioning slows the outer loop—an intermediate G is the cheaper per-iteration operating point (Fig. 5a), and its group rank is a dial on the trade-off. (iii) *Channels.* By the duality, adding channels to a single-channel dictionary raises the update rank exactly as adding images does: cost tracks $N_b=CK$ regardless of the split (11.55 to 12.48 ms at $N_b=12$ across $C \in \{1, 2, 3\}$, Fig. 5c), so a solver chosen for large- K dictionary updates is the right choice for large- C ones and, dually, for large- C sparse coding. (iv) *Channel correlation.* An mc-dict model shares one code across channels and suits strongly correlated channels; an sc-dict model gives each channel its own code and suits weakly correlated ones; the update cost of the two is set by their respective N_b (K vs. CK).

9 Where the abstraction is incomplete

Forward-backward is a limit, not an ADMM instance. The FISTA corner is the majorised limit (7) of the ADMM branch, not a point of the partition; the two branches share the master problem and the constraint prox but not the inner iteration. We therefore present the driver as one operator-splitting family with two branches sharing a derivation, not as a single ADMM recursion covering FISTA. Recovering the FISTA reference exactly (Table 2) required matching its fixed step $L=14K$ and Nesterov schedule, which the driver exposes as the acceleration coordinate.

The dual initialisation is a per-family convention. SPORCO seeds the scaled dual differently for its equality and consensus classes ($\mathbf{u}^0=\mathbf{g}^0$ vs. $\mathbf{g}^0/\rho$). Reproducing a reference *exactly* requires adopting its seed, so we hold this convention to each endpoint. It is a transient, not a modelling choice: on the convex dictionary-update subproblem (codes fixed), the equality, consensus, and zero initialisations converge to the same filters to 3.9×10^{-7} over 300 inner iterations, even though their first iterates differ. This is why the same interior driver collapses onto both endpoints (§6) despite their differing seeds.

Non-convex bases diverge in path. The block-DFT equivalence (§7) is Parseval-exact for the objective but, because C_{PN} is non-convex, the spatial and block-DFT consensus follow different

paths to possibly different stationary points; the equivalence is of the problem class, matching the scope of the original “3D = consensus” argument. *Not covered.* Two elements of the reviewed study lie outside the driver as stated: the spatial-mask (inpainting) updates add a second variable split that our partition coordinate does not express, and the inter-channel joint-sparsity model lives in the sparse-coding subproblem, which we hold fixed; both are natural extensions (§10) but are not claimed here.

10 Related work and positioning

Our object is the comparative review of convolutional dictionary updates (Garcia-Cardona and Wohlberg, 2018a) and the algorithms it consolidates: the equality-constrained ADMM with ISM/CG inner solves (Wohlberg, 2016b), the consensus update (Šorel and Šroubek, 2016), and the FISTA update (Beck and Teboulle, 2009), coupled by the single-inner-iteration alternation of (Garcia-Cardona and Wohlberg, 2017) and implemented in SPORCO (Wohlberg, 2017). That study already carries the updates to C channels in its Sec. VI, names both channel representations, states the $K \leftrightarrow C$ duality, and runs colour experiments (Wohlberg, 2016a); subsequent work develops acceleration and convergence (Garcia-Cardona and Wohlberg, 2018b) and online CDL (Liu et al., 2018; Mairal et al., 2010). Reproducing the multi-channel updates on colour is thus *baseline work*, and we treat it as such (§5). Our contribution is not that the updates run on colour; it is the single grouped-consensus derivation from which the grayscale updates, both channel models, and the methods between the endpoints all follow by setting a few coordinates, together with the interior cells that no prior work instantiates and the equivalences that the settings make algebraic. Consensus and proximal splitting are classical (Boyd et al., 2011; Parikh and Boyd, 2014; Combettes and Pesquet, 2011); the grouped-consensus interior and its use as a cost/conditioning dial for the convolutional dictionary update are, to our knowledge, new. Convolutional sparse coding underlies applications from image restoration to hyperspectral analysis (Heide et al., 2015; Pappan et al., 2017; Zhang et al., 2014), for which the multi-channel selection map (§8) is directly relevant.

11 Conclusion

The convolutional dictionary update is usually met as a menu of separately derived algorithms. We showed it is one grouped-consensus splitting: the equality and consensus updates are the two ends of a single partition, the CG and exact inner solves and the FISTA update are further settings of the same driver, and the whole construction carries verbatim to multi-channel signals. Instantiating the driver reproduces the reference implementations to machine precision; its interior—filter updates with intermediate group counts—supplies working methods that no released solver provides and that collapse onto the references at the extremes; and three known equivalences, including the multi-channel cost identity and the image/channel duality, drop out as algebra of the settings. What one derivation buys is not a faster benchmark but a map: which update to run for which image-count, channel-count, and channel-correlation regime, with the reason read off the structure of the per-frequency system. The mask and joint-sparsity extensions, and a learned choice of group rank, are natural next steps.

Reproducibility. Every number traces to a logged run; the driver, the validation harness that runs SPORCO and compares, the multi-channel and interior instances, and the figure/number generators are provided under `proposal/`. The framework and its coordinates were registered before the validation runs.

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